BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Binary Addition and Shifts

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. Convert 00101101 binary to denary.

2. What is a character set?
   A) Characters linked to codes
   B) A list of fonts
   C) A list of filenames
   D) A collection of pixel colours

3. "A record must contain only one unnamed value." Correct this. Explain why.

4. Valid ages are 13-18. Changing <18 to <=18 fixes a bug. Give three tests near 18 and their expected results. Why retest a value that worked before?

Today's objectives

Recall key facts about binary addition and shifts. Work through a new example with correct technical language.

Key vocabulary - Read, cover, recall

Carry
Value passed to the next binary column during addition.

Overflow
Result exceeds the largest value representable in fixed-width storage.

Logical left shift
Move bits left and insert zeros on right.

Logical right shift
Move bits right and insert zeros on left.

BLACK PEN: Write one definition in your own words.

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New Learning - Key ideas

Binary addition rules

0+0=0; 0+1=1; 1+1=10, so write 0 and carry 1 into the next column.

Work right to left

Line up bits, add each column including any carry, then record the output.

Logical shifts

A logical left or right shift moves each bit by the stated number of places, introducing zeros at the vacated end.

Overflow and loss

With fixed eight-bit storage, a carry beyond bit 7 is lost (overflow); bits shifted out are discarded.

I Do - Worked model

Add the eight-bit values 00010110 and 00001101, showing the carry where needed. [4 marks]

- Line up eight positions
- Add from the right: 0+1=1; next 1+0=1.
- At the third bit, 1+1=0 carry 1; continue including the carry.
- The eight-bit result is 00100011 (35 denary). [4]

We Do - Guided practice

1. Calculate 00101101 + 00000111 in binary. [4]

2. Perform a logical left shift of 00010101 by one place. [3]

3. Perform a logical right shift of 10110000 by two places. [3]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Change the multiplier to model a three-place logical left shift without overflow.

```
value = 9
result = value * 2
print(result)
```

You Do - Independent exam questions

Q2. Add 00011001 and 00000111 as eight-bit unsigned binary. [3]

Q3. Apply a logical left shift of two places to 00001011. [2]

Q4. Explain why adding 11111111 to 00000001 causes overflow in an eight-bit register. [3]

Exit Ticket - True or False?

An algorithm only gives reliable results when its inputs are handled as specified.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. 45, because 32 + 8 + 4 + 1 = 45.

2. A - Characters linked to codes

3. Named fields about one item or person. Fields may have different data types.

4. 17 accepted; 18 accepted; 19 rejected. Retest a value that worked before to check for regression: a new fault caused by the fix.

I Do - Explain the model

Line up eight positions

Add from the right: 0+1=1; next 1+0=1.

At the third bit, 1+1=0 carry 1; continue including the carry.

The eight-bit result is 00100011 (35 denary). [4]

We Do - Model answers

1. 00101101 (45) + 00000111 (7) = 00110100 (52) [4; credit aligned working/carries].

2. 00101010 [2]; all bits move one position left and 0 enters at the right [1].

3. 00101100 [2]; two zeros enter on the left [1].

Code First - Answer

```
value = 9
result = value * 8
print(result)
```

Expected output: 72

You Do - Self-mark

Q2: 00011001 (25) + 00000111 (7) = 00100000 (32) [3].

Q3: 00101100 [2].

Q4: The mathematical result is 256 [1], requiring a ninth bit [1]; the carry beyond eight bits cannot be stored [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.